

TANUSH THAKRAN

AI / ML Engineer

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SUMMARY

Machine learning engineer who ships models with evidence behind them: leakage-safe evaluation, controlled ablations, and metrics reported against honest baselines. Experience spans learning-to-rank on multi-million-row datasets, applied ML with scikit-learn, and LLM/RAG systems, owning the full path from data extraction through deployment.

TECHNICAL SKILLS

Machine Learning: LightGBM, scikit-learn, Feature Engineering, Cross-Validation, Model Evaluation. **Familiar with:** PyTorch, TensorFlow
LLM & Retrieval: RAG, Embeddings, Vector Search (ChromaDB), Prompt Engineering, LLM Evaluation, Groq API, Gemini API, Whisper, Ollama
Data & Analytics: Python, SQL, PostgreSQL, Pandas, NumPy, PyArrow/Parquet, Power BI, DAX, Power Query
Web & Infra: FastAPI, Node.js, React.js, Next.js, Celery, Redis, Docker, Streamlit, pytest, GitHub Actions, Git, Flutter

MACHINE LEARNING & DATA PROJECTS

Pokémon TCG Move Prediction — Learning-to-Rank

Python, LightGBM, PyArrow | github.com/Tanush1206/pokemon-tcg-move-prediction

- Framed move selection as learning-to-rank: an LGBMRanker with a LambdaRank objective trained on 3.35M rows (83 features) from 4,604 recorded battles, reaching 57.4% top-1 accuracy on 50,649 held-out decisions vs. a 21.7% random baseline (2.65x); the true move is in the top 3 88% of the time.
- Ran a 7-run ablation on a fixed test set with 70/15/15 episode-level splits: the largest gain (+0.054) came from reverse-engineering undocumented option indices to resolve which card each option refers to, while filtering for “stronger” episodes did not help (matched-volume control).
- Built a multiprocessing extraction pipeline: 4,604 game files streamed into one Parquet file in about 75 seconds on 31 workers.

Customer Churn Prediction — Deployed ML App

Python, scikit-learn, Streamlit | github.com/Tanush1206/Customer-Churn-Prediction

- Built an end-to-end pipeline on the IBM Telco dataset (7,043 customers): leakage-safe ColumnTransformer; logistic regression, random forest and gradient boosting compared on held-out data; logistic regression kept for interpretability (0.003 ROC-AUC below gradient boosting) and tuned with 5-fold CV. Chose a 0.30 threshold from out-of-fold predictions, lifting churn recall from 55.9% to 75.1% at 51.8% precision (84.1% ROC-AUC); deployed as a Streamlit app with per-feature contributions.

RAG-Based AI — Offline Retrieval-Augmented QA Pipeline

Python, Whisper, Ollama | github.com/Tanush1206/rag_based_ai

- Built a fully offline RAG pipeline over Hindi-language course videos (ffmpeg, Whisper large-v2 transcription with translation to English, local bge-m3 embeddings, top-5 cosine retrieval, deepseek-r1 via Ollama); answers cite video, title and timestamp and say so when the answer is missing; zero external API calls; fixed a scale-only embedding crash with batched requests.

LLM Utility Lab

Python, Groq API, FastAPI, Docker | github.com/Tanush1206/llm-utility-lab

- Built a modular LLM service on the Groq API (FastAPI, Docker, deployed on Render) with per-request token tracking, typed error handling, and an evaluation runner that records model, temperature and prompt version for each run and flags pass-to-fail regressions; 58 pytest tests run in GitHub Actions CI.

Superstore Profitability Analysis — Power BI

Power BI, Power Query, DAX | github.com/Tanush1206/superstore-powerbi-analysis

- Analysed 9,994 line items (\$2.3M revenue, 2014–2017): margin is 29.5% undiscounted, turns negative past a 25% discount and reaches –77% beyond 40%. Central region discounts 24.0% for a 7.9% margin vs. West's 10.9% for 14.9%, a gap worth about \$35K over four years; proposed a single-region test before rollout; reconciled the data load against control totals.

VideoCaptionMaker — AI Video Captioning App (Feature-Complete, Pre-Deployment)

Python, FastAPI, Celery, ChromaDB, Next.js | github.com/Tanush1206/Video-Caption-Maker

- Built a six-service app (FastAPI, Next.js 14, Celery worker running faster-whisper large-v3 on GPU, Redis, PostgreSQL, ChromaDB) with a caption editor, translation, FFmpeg burn-in export, and Gemini-backed Q&A that declines when retrieval finds nothing relevant.
- Secured with httpOnly-cookie JWT refresh tokens, bcrypt, rate limiting and upload validation; 250+ backend tests.

EXPERIENCE

App Developer Intern — CYPARTA

Dec 2025 – Jan 2026

Cairo, Egypt (Remote) | Flutter, Firebase

- Built the Driver & Medic mobile app end-to-end, handling real-time GPS telemetry and state consistency across concurrent users for live logistics.
- Implemented trip-lifecycle workflows and post-trip EMS reporting modules for data integrity.

Junior Manager — AIESEC

Oct 2025 – Mar 2026

Bengaluru, India | Managed Global Volunteer operations, tracked departmental KPIs, led cross-functional teams.

EDUCATION

Integrated B.Sc. in Computer Science — BITS Pilani

2024 – 2028

CGPA: 7.64 | Coursework: Algorithms, Data Structures, Databases, Operating Systems, Networks, System Design, Statistics, Deep Learning

CERTIFICATIONS

Machine Learning: Advanced Learning Algorithms — Stanford / DeepLearning.AI • Supervised ML: Regression & Classification — Stanford / DeepLearning.AI • Intermediate Machine Learning — Kaggle

LLM & GenAI: LangChain: Chat with Your Data (LangChain, DeepLearning.AI) • Building Systems with the ChatGPT API (OpenAI, DeepLearning.AI) • ChatGPT Prompt Engineering for Developers (OpenAI, DeepLearning.AI) • AI Fluency: Framework & Foundations (Anthropic)